Exercise - 1:

Complete all the Task.

1. Read and display the image.

• Read the image using the Pillow library and display it.

• You can also use matplotlib to display the image.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
from PIL import Image
import matplotlib.pyplot as plt
import numpy as np


image = Image.open("/content/drive/MyDrive/AI and Machine Learning/WEEK1/images.jpg")


plt.imshow(image)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image

2. Display only the top left corner of 100x100 pixels.

• Extract the top-left corner of the image (100x100 pixels) and display it using NumPy and Array Indexing.

```
image_array = np.array(image)

top_left = image_array[0:100, 0:100]

plt.imshow(top_left)
plt.title("Top Left 100x100 Pixels")
plt.axis("off")
plt.show()
```

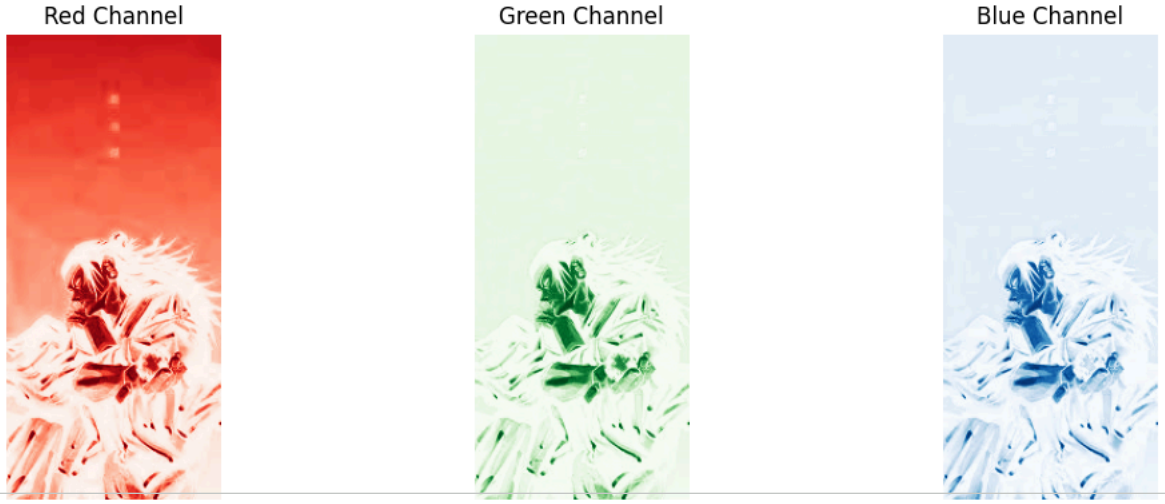
Top Left 100x100 Pixels

3. Show the three color channels (R, G, B).

• Separate the image into its three color channels (Red, Green, and Blue) and display them individually, labeling each channel as R, G, and B.{Using NumPy.}

```
red_channel = image_array[:, :, 0]
green_channel = image_array[:, :, 1]
blue_channel = image_array[:, :, 2]

plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(red_channel, cmap="Reds")
plt.title("Red Channel")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(green_channel, cmap="Greens")
plt.title("Green Channel")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(blue_channel, cmap="Blues")
plt.title("Blue Channel")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Red Channel Green Channel Blue Channel

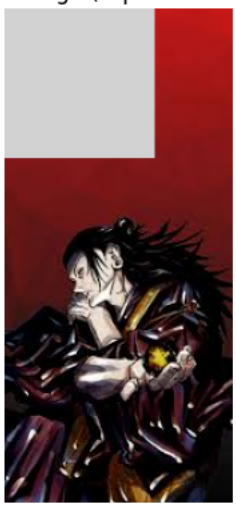


4. Modify the top 100 × 100 pixels to a value of 210 and display the resulting image:

• Modify the pixel values of the top-left 100 × 100 region to have a value of 210 (which is a light gray color), and then display the modified image.

```
modified_image = image_array.copy()

modified_image[0:100, 0:100] = 210

plt.imshow(modified_image)
plt.title("Modified Image (Top 100x100 = 210)")
plt.axis("off")
plt.show()
```

Modified Image (Top 100x100 = 210)



Exercise - 2:

Complete all the Task.

1. Load and display a grayscale image. • Load a grayscale image using the Pillow library. • Display the grayscale image using matplotlib.

```
image_grayed = Image.open("/content/drive/MyDrive/AI and Machine Learning/WEEK1/images.jpg").convert("L")

image_array_grayed = np.array(image_grayed)

plt.imshow(image_array_grayed, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```

Grayscale Image

2. Extract and display the middle section of the image (150 pixels).

• Extract a 150 pixel section from the center of the image using NumPy array slicing. • Display this cropped image using matplotlib.

```
height, width = image_array_grayed.shape

center_h = height // 2
center_w = width // 2

middle_section = image_array_grayed[
    center_h-75:center_h+75,
    center_w-75:center_w+75
]

plt.imshow(middle_section, cmap="gray")
plt.title("Middle 150x150 Section")
plt.axis("off")
plt.show()
```

Middle 150x150 Section

3. Apply a simple threshold to the image (e.g., set all pixel values below 100 to 0).

• Apply a threshold to the grayscale image: set all pixel values below 100 to 0, and all values above 100 to 255 (creating a binary image).

• Display the resulting binary image.

```
binary_image = np.where(image_array_grayed < 100, 0, 255)

plt.imshow(binary_image, cmap="gray")
plt.title("Binary Image (Threshold = 100)")
plt.axis("off")
plt.show()
```

Binary Image (Threshold = 100)

4. Rotate the image 90 degrees clockwise and display the result.

- Rotate the image by 90 degrees clockwise using the Pillow rotate method or by manipulating the image array.
- Display the rotated image using matplotlib.

```
rotated_image = image_grayed.rotate(-90, expand=True)

plt.imshow(rotated_image, cmap="gray")
plt.title("Rotated 90 Degrees Clockwise")
plt.axis("off")
plt.show()
```

Rotated 90 Degrees Clockwise

5. Convert the grayscale image to an RGB image.

- Convert the grayscale image into an RGB image where the grayscale values are replicated across all three channels (R, G, and B).
- Display the converted RGB image using matplotlib.

```
rgb_image = image_grayed.convert("RGB")

plt.imshow(rgb_image)
plt.title("Grayscale Converted to RGB")
plt.axis("off")
plt.show()
```

Grayscale Converted to RGB

3. Image Compression and Decompression using PCA.

1. Load and Prepare Data:

- Fetch an image of you choice.{If colour convert to grayscale}

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
```

```
image = Image.open("/content/drive/MyDrive/AI and Machine Learning/WEEK1/images.jpg").convert("L")

image_array = np.array(image)

height, width = image_array.shape

data = image_array.reshape(height, width)

plt.imshow(image_array, cmap="gray")
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image

- Center the dataset - Standaridze the Data.

```
mean = np.mean(data, axis=0)
centered_data = data - mean
```

- Calculate the covaraince matrix of the Standaridze data.

```
cov_matrix = np.cov(centered_data, rowvar=False)

print("Covariance Matrix Shape:", cov_matrix.shape)
```

```
Covariance Matrix Shape: (152, 152)
```

2. Eigen Decomposition and Identifying Principal Components:

- Compute Eigen Values and Eigen Vectors.

```
eigenvalues, eigenvectors = np.linalg.eigh(cov_matrix)

print("Eigenvalues Shape:", eigenvalues.shape)
print("Eigenvectors Shape:", eigenvectors.shape)
```

```
Eigenvalues Shape: (152,)
Eigenvectors Shape: (152, 152)
```

- Sort the eigenvalues in descending order and choose the top k eigenvectors corresponding to the highest eigenvalues.

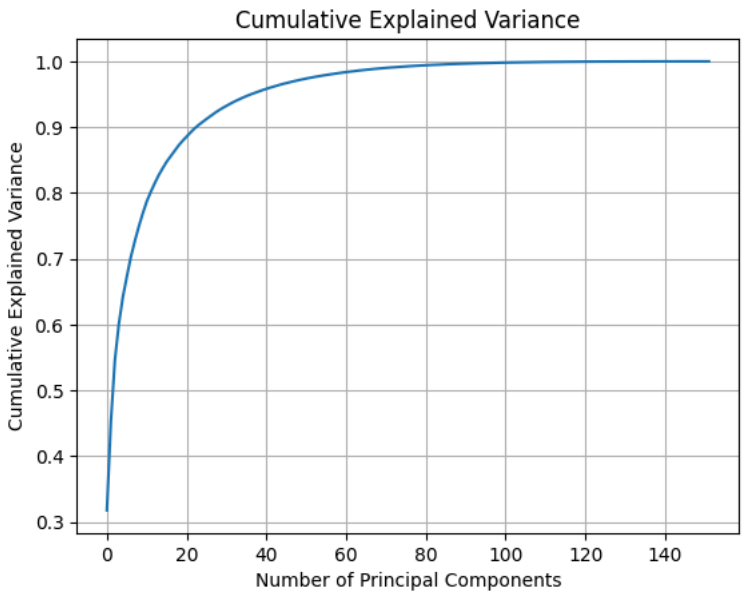
```
sorted_indices = np.argsort(eigenvalues)[::-1]

eigenvalues = eigenvalues[sorted_indices]
eigenvectors = eigenvectors[:, sorted_indices]


k = 50
top_k_eigenvectors = eigenvectors[:, :k]
```

```
print("Top k Eigenvectors Shape:", top_k_eigenvectors.shape)
```

```
Top k Eigenvectors Shape: (152, 50)
```

Identify the Principal Components with the help of cumulative Sum plot.

```
explained_variance_ratio = eigenvalues / np.sum(eigenvalues)

cumulative_variance = np.cumsum(explained_variance_ratio)

plt.plot(cumulative_variance)
plt.title("Cumulative Explained Variance")
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance")
plt.grid(True)
plt.show()
```



3. Reconstruction and Experiment:

• Reconstruction: Transform the original data by multiplying it with the selected eigenvec- tors(PCs) to obtain a lower-dimensional representation.

```
compressed_data = np.dot(centered_data, top_k_eigenvectors)

print("Compressed Data Shape:", compressed_data.shape)
```

```
Compressed Data Shape: (330, 50)
```

```
reconstructed_image = np.dot(compressed_data, top_k_eigenvectors.T) + mean

print("Reconstructed Image Shape:", reconstructed_image.shape)
```

```
Reconstructed Image Shape: (330, 152)
```

```
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(image_array, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(reconstructed_image, cmap="gray")
plt.title(f"Reconstructed Image (k={k})")
plt.axis("off")

plt.show()
```

- Experiments: Pick Four different combination of principal components with various ex- plained variance value and compare the result.

```
k_values = [10, 30, 60, 120]

plt.figure(figsize=(10,8))

for i, k in enumerate(k_values):
    components = eigenvectors[:, :k]
    compressed = np.dot(centered_data, components)
    reconstructed = np.dot(compressed, components.T) + mean

    plt.subplot(2,2,i+1)
    plt.imshow(reconstructed, cmap="gray")
    plt.title(f"k = {k}")
    plt.axis("off")

plt.tight_layout()
plt.show()
```

k = 10

k = 30